ROBT613 · MIDTERM PROJECT

Can a P300 decoder skip calibration?

Cross-subject transfer of a shrinkage-LDA ERP decoder on BrainInvaders2013a

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17
subjects, IDs 8–24

34
held-out test runs

7.4 min
fresh Colab CPU run

RESEARCH QUESTION

Can other users' data replace calibration?

Every P300 session normally starts with a calibration phase. If a decoder trained on other people works out of the box, a new user can start immediately.

We measure how much an in-scope linear decoder loses when it skips calibration, and for whom.

H1

Transfer works

A leave-one-subject-out (LOSO) decoder ranks targets above chance (AUC > 0.5).

H2

Calibration still helps

Within-subject training beats LOSO on the same held-out run.

H3

Not just more data

Within still beats LOSO when training class counts are matched (LOSO-matched).

Subjects 8–10 informed development in an earlier pilot; a sensitivity analysis excludes them (n = 14).

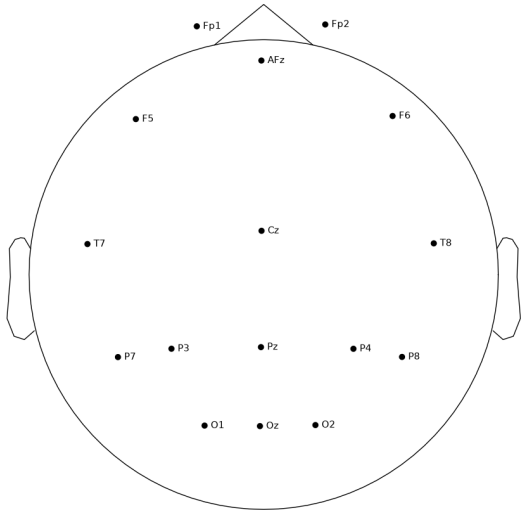
Data audit and provenance

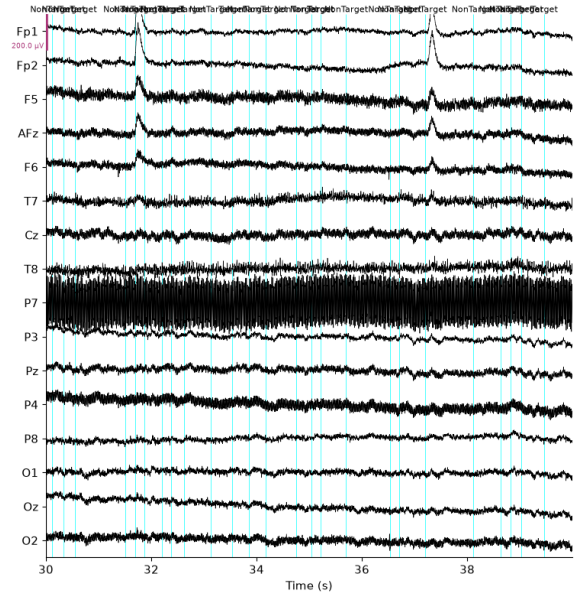
17
subjects (IDs 8–24), one session each

16
EEG channels at 512 Hz, left-earlobe reference

34
training runs in total, two per subject

80 / 400
targets / non-targets per run (1 : 5)

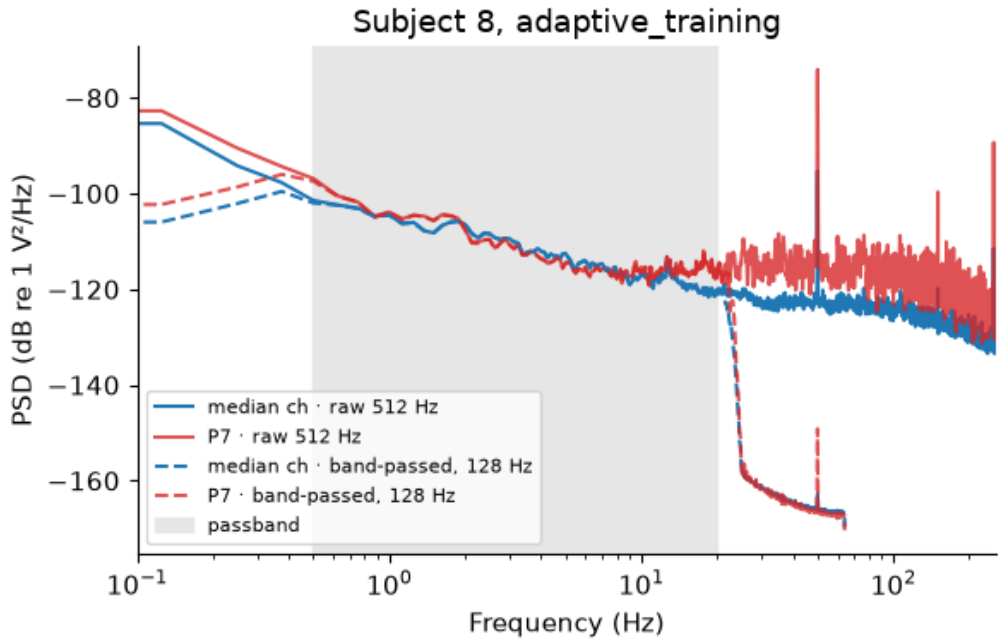


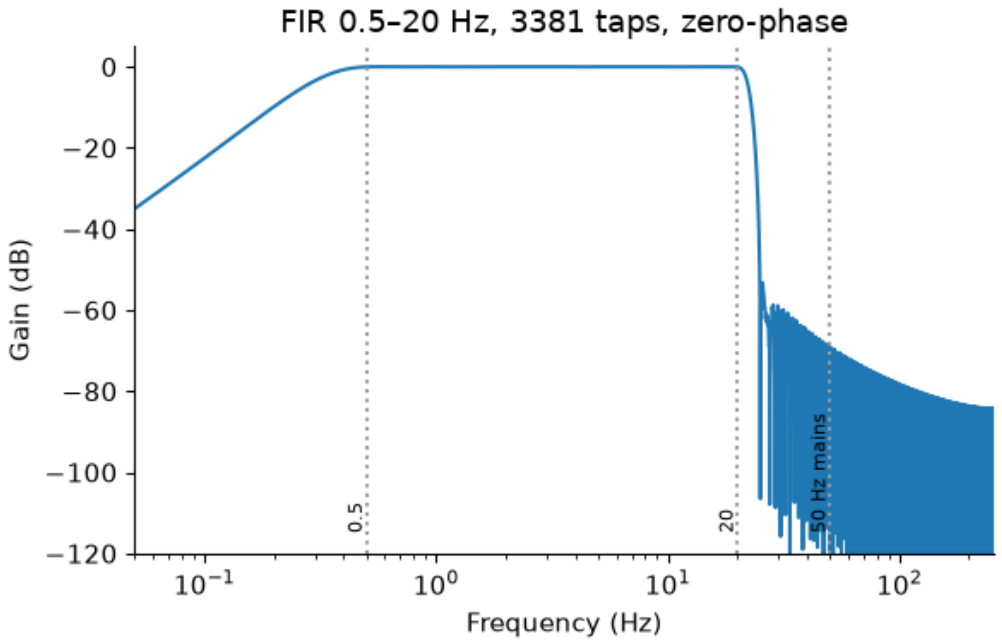


Loaded from the original Zenodo files through MOABB into MNE Raw objects. Every run is checked by assertions before analysis.

Left: sensor layout from the MOABB/MNE montage. Right: raw EEG excerpt with flash events.

Filter choice backed by the spectrum





0.5–20 Hz

FIR band-pass, zero-phase, fitted per run

128 Hz

resampled from 512 Hz together with the events

No notch

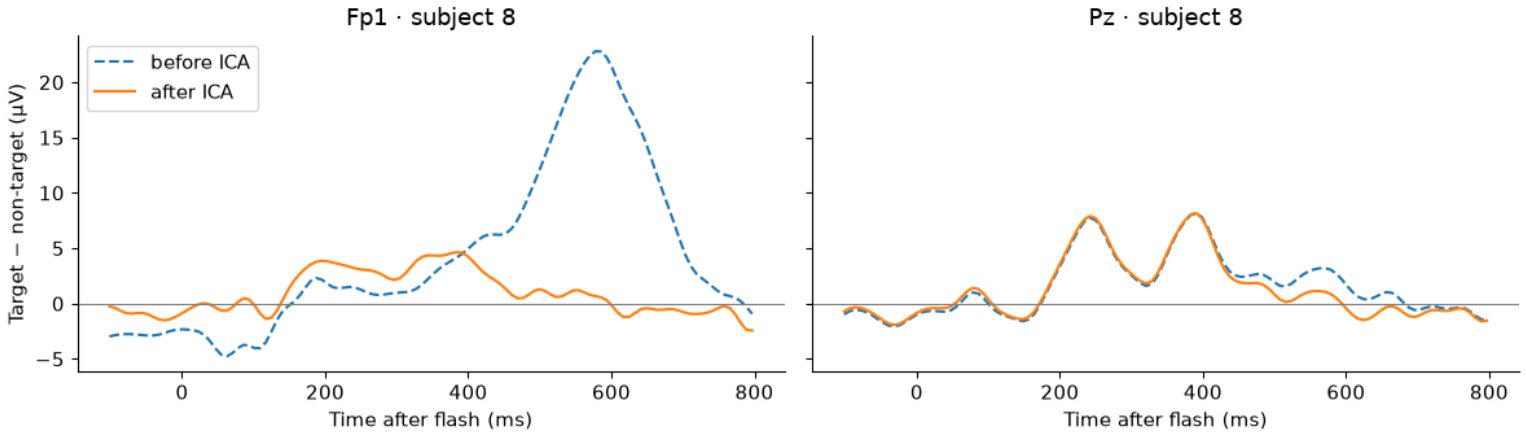
50 Hz is already about −75 dB after the low-pass

3.3 s

edge guard equal to the filter half-support

Subject 8: PSD before and after band-pass (left) and the FIR frequency response (right).

Artifact handling and quality control



1. FastICA, 15 components, fitted on a 1–20 Hz copy of each unlabelled run
2. A component is removed only if the Fp1/Fp2 proxy correlation and the frontal map agree
3. Peak-to-peak rejection at 100 µV on the cleaned epochs (threshold sweep in notebook)

97.4%

of epochs retained across all 34 runs

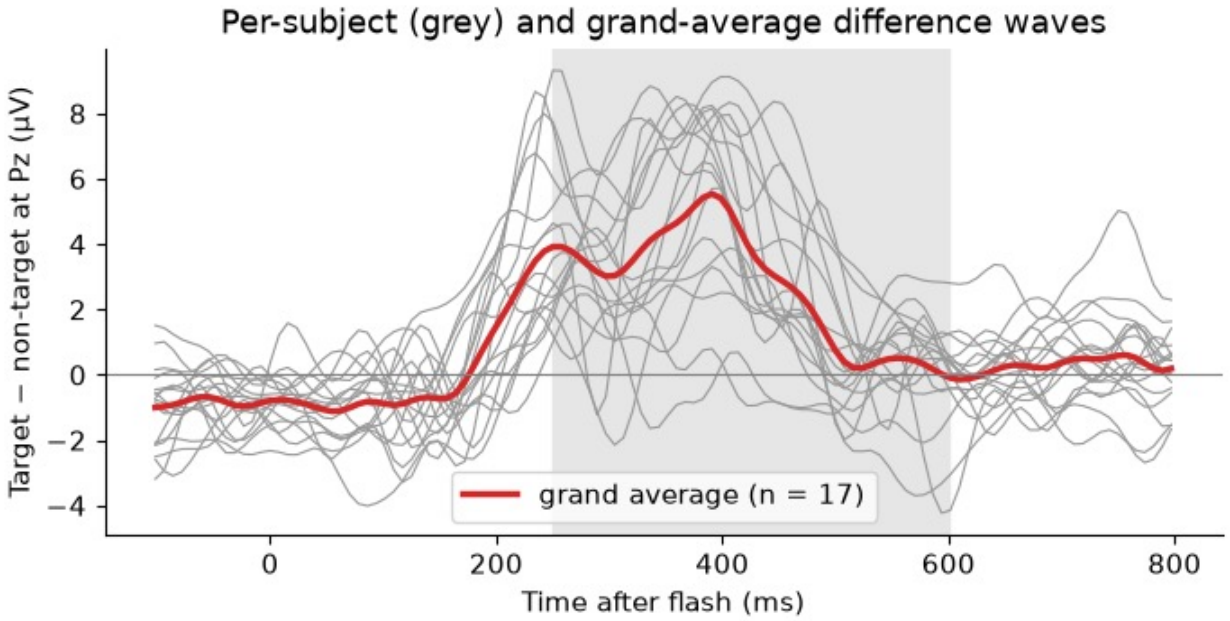
Edge drops
3 target / 12 non-target

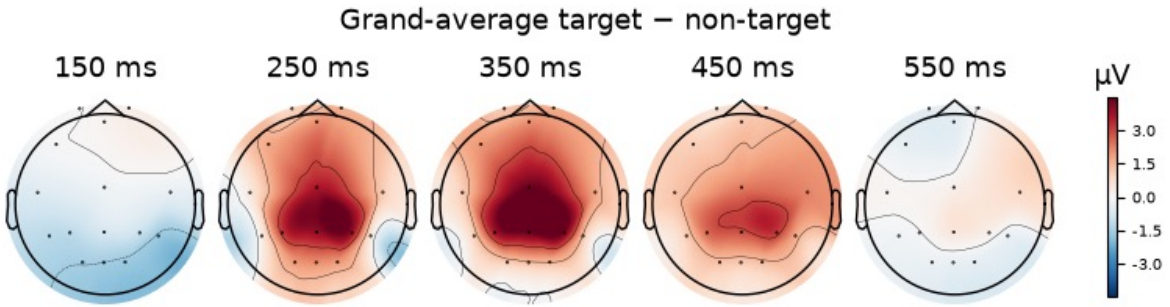
Amplitude drops
80 target / 329 non-target

Bad channel
F6, outside the decoder montage

Subject 8, target − non-target at Fp1 and Pz before (dashed) and after ICA.

The P300 is there before we decode it





16 / 17

subjects with a positive target – non-target mean at Pz

2.74 µV

mean Pz contrast in the predeclared window

Window 250–600 ms fixed in advance (notebook 01). It is never tuned on decoding accuracy.

One leakage-free pipeline

Epoch

0–800 ms after each flash, 128 Hz

ERP features

16 × 50 ms window means at Cz, P3, Pz, P4, Oz = 80

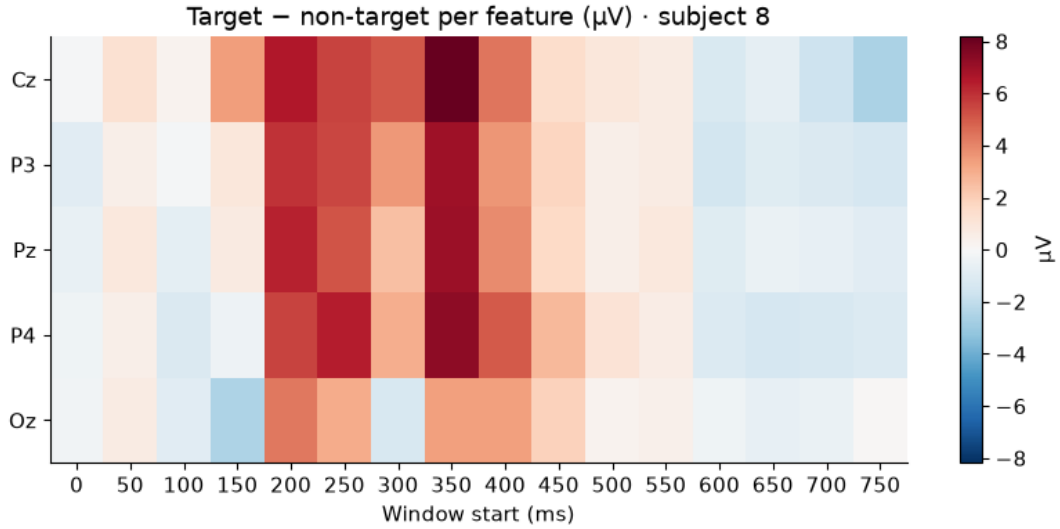
StandardScaler

mean and SD from the training fold

Shrinkage LDA

analytic (Ledoit–Wolf) shrinkage

fitted on training folds only



1. One scikit-learn Pipeline, as in notebook 02
2. No hyperparameters tuned, so nothing is chosen on test data
3. Montage and windows frozen before the full-cohort run
4. Same pipeline in all three conditions

Same test epochs, different training data

Within

TRAINS ON

the subject's own other run

TESTS ON

the same held-out run

Leave-one-run-out, 2 folds

LOSO-matched

TRAINS ON

other 16 users, subsampled to the Within class counts

TESTS ON

the same held-out run

10 seeded draws averaged

LOSO

TRAINS ON

all runs of the other 16 users

TESTS ON

the same held-out run

Test subject asserted absent

Metric

ROC AUC (primary) and balanced accuracy; one score per subject

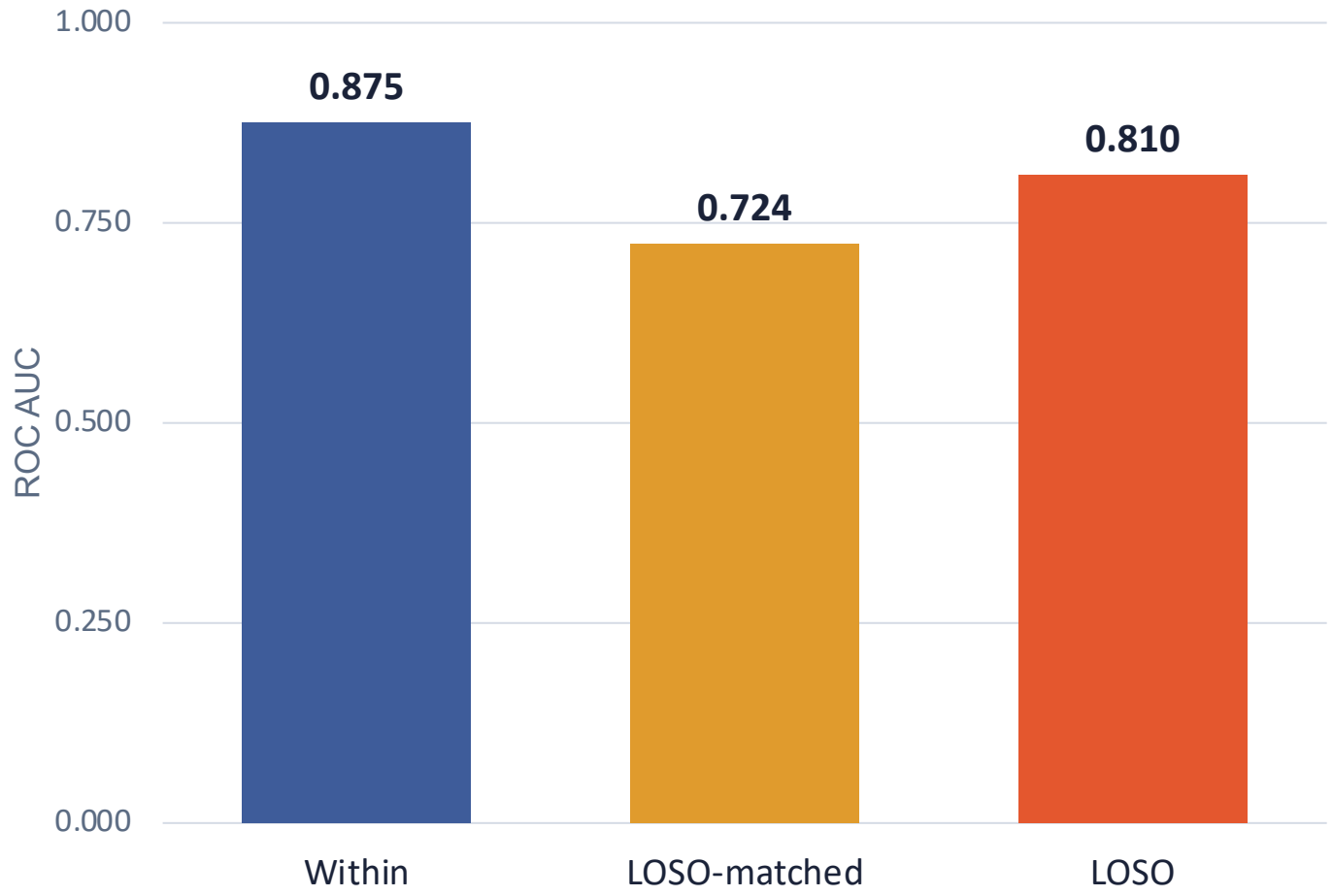
Chance level

Within: 200 within-run label permutations; LOSO: 1,000 test-label shuffles

Variability

mean ± SD across subjects and across test folds

Transfer works, calibration still wins



Within

0.875 ± 0.056

LOSO-matched

0.724 ± 0.071

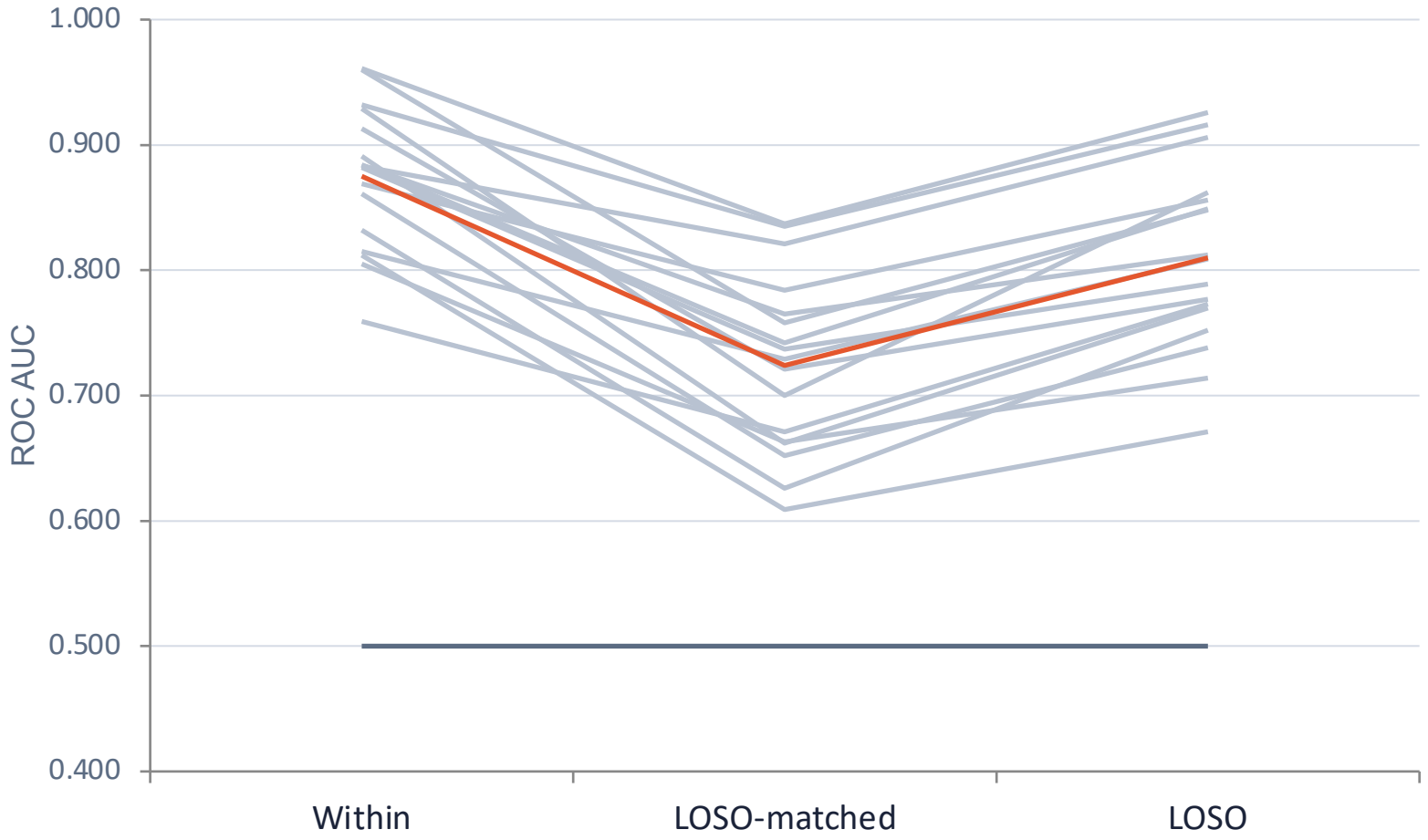
LOSO

0.810 ± 0.072

34 / 34 LOSO test runs beat their permutation null (p < .05)

Mean ± SD across 17 subjects; each subject = mean of 2 held-out runs. Chance AUC = 0.5.

Every user transfers, but not equally



15 / 17

users score higher with their own calibration

0.072

median Within − LOSO gap

0.018

median Within fold SD (2 folds)

0.671

lowest LOSO AUC, still above chance

Grey: individual subjects. Coral: cohort mean. Slate: chance (0.5).

All three hypotheses hold in this cohort

H1 LOSO > chance

+0.309

median AUC difference

Holm p = 2.3 × 10^{-5}

rank-biserial r = 1.00

H2 Within > LOSO

+0.072

median AUC difference

Holm p = 1.9 × 10^{-4}

rank-biserial r = 0.90

H3 Within > LOSO-matched

+0.142

median AUC difference

Holm p = 2.3 × 10^{-5}

rank-biserial r = 1.00

Sensitivity without pilot subjects 8–10 (n = 14): Holm p = 1.8 × 10^{-4} / 3.1 × 10^{-4} / 1.8 × 10^{-4} for H1 / H2 / H3. Same conclusions.

Paired Wilcoxon signed-rank tests across 17 subjects, Holm-corrected for the three primary hypotheses.

CONCLUSION

What the result shows

A decoder trained only on other users works on a new user (AUC 0.810), but personal calibration still adds 0.072 AUC (median).

More external data helps: matching the training size lowers transfer AUC from 0.810 to 0.724.

LIMITS

1. Offline: zero-phase filters and per-run ICA use future samples
2. Single-flash AUC is not online command accuracy
3. One protocol, 17 single-session users
4. LOSO models share training users; p-values summarise this cohort

BACKUP

References and AI disclosure

1. E. Vaineau et al., “Brain Invaders adaptive versus non-adaptive P300 BCI dataset,” arXiv:1904.09111, 2019

2. A. Barachant, M. Congedo, “A Plug&Play P300 BCI using information geometry,” arXiv:1409.0107, 2014

3. J. Kim, S.-P. Kim, “A Plug-and-Play P300-based BCI with zero-training application,” IEEE TNSRE, vol. 33, 2025

4. V. Jayaram, A. Barachant, “MOABB: trustworthy algorithm benchmarking for BCIs,” J. Neural Eng., 2018

5. B. Abibullaev, “ROBT613 BCI Workshop,” notebooks 00–06, github.com/berdakh/ROBT613-BCI-Workshop

AI TOOLS

OpenAI Codex
code review, debugging, API lookup, implementation, statistics checks, plotting, text drafting

Claude Code
requirements check, wording edits, slide generator

No AI tool generated results, citations or dataset facts. Every number comes from the executed notebook.

BACKUP

Reproducibility record

Package	Version
mne	1.13.2
moabb	1.7.2
numpy	2.5.4
scipy	1.18.1
pandas	2.3.3
scikit-learn	1.9.1
matplotlib	3.11.2
PyYAML	6.0.3
remotezip	0.12.6
threadpoolctl	3.7.0

7.41 min

fresh Colab CPU run with an empty data cache

613

random seed for ICA, permutations and draws

17

subjects, IDs 8–24 listed in the manifest

0

errors across all 32 code cells

Shared code lives in functions; the notebook runs top to bottom with versions pinned.